Human-Assisted Cognitive Defect Inspector (HCDI)

- AI-Powered Visual Defect Detection System
  - Artificial Intelligence Capstone Project
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Problem Statement

- Manual inspection is time-consuming and inconsistent.
  - Human fatigue can reduce defect detection accuracy.
  - Need for automated intelligent inspection systems.

Project Objective

- Develop CNN-based defect classification model.
  - Estimate defect severity quantitatively.
  - Generate human-readable AI inspection remarks.

System Architecture

- Image Input → Preprocessing → CNN Feature Extraction
  - Binary Classification (Defect / No Defect)
  - Severity Estimation & AI Remark Generation

Technology Stack

- Python
  - Google Colab
  - TensorFlow / Keras
  - OpenCV
  - NumPy & Matplotlib

Unique Dataset Creation

- Synthetic dataset generated programmatically.
  - Defect samples contain artificial irregular regions.
  - Ensures originality and avoids external dataset usage.

Image Preprocessing

- Resizing images to 128×128 pixels.
  - Normalization using rescale = 1/255.
  - Training-validation split with ImageDataGenerator.

CNN Model Design

- Convolution layers extract spatial features.
  - MaxPooling layers reduce dimensions.
  - Dense layers perform final binary classification.

Model Training Strategy

- Binary Cross-Entropy loss function.
  - Adam optimizer for faster convergence.
  - Model trained over multiple epochs.

Inspection Intelligence Engine

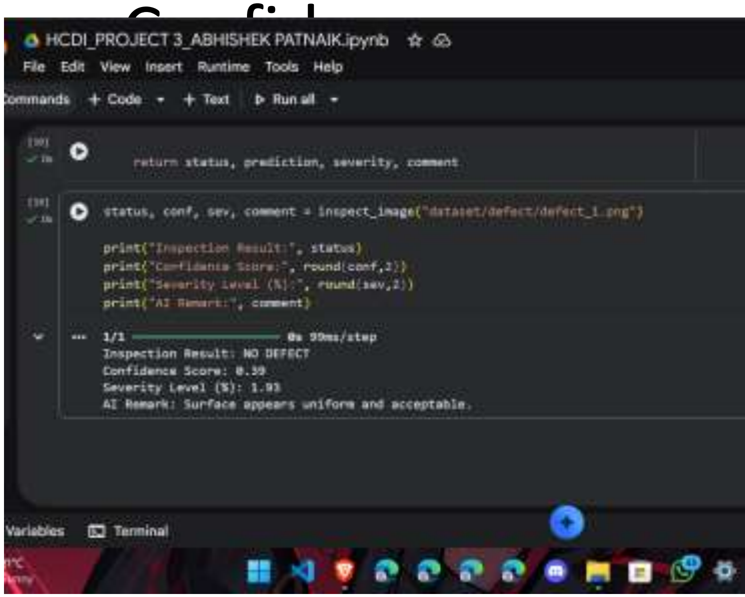
- Predicts probability of defect presence.
  - Calculates confidence score.
  - Computes severity level using pixel intensity ratio.

Severity Estimation Mechanism

- Dark pixel concentration analysis.
  - Higher irregular region → Higher severity score.
  - Provides quantitative defect percentage.

Sample Inspection Output

- Model outputs classification result,
  …verity percentage, and AI



Project Implementation Link

- Google Colab Notebook:
  - https://colab.research.google.com/drive/12uFYvPwUzOlD_urd5LEO7TZ_8Tjv5fhl?usp=sharing

Applications

- Manufacturing Quality Control
  - Automotive Component Inspection
  - Electronics PCB Testing
  - Industrial Automation Systems

Conclusion & Future Scope

- Automates industrial inspection tasks efficiently.
  - Reduces human error in quality evaluation.
  - Future: Integrate Grad-CAM explainability & deployment.